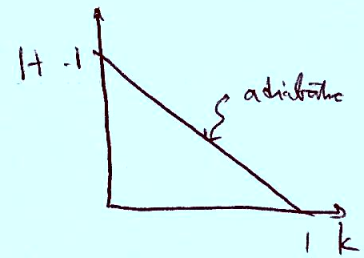
2/5/19

Lecture 7

Let $\Phi \equiv \dfrac{S_a}{(C_p T_{ti})^{1/2}} = \dfrac{T}{\dot{m}\sqrt{C_p T_{ti}}}$



Can show that

$$\Phi = \frac{\sqrt{C_p T_{ti}}}{u}\left(\frac{2u^2}{2C_p T_{ti}} + \frac{(\gamma-1)}{\gamma}\frac{C_p T}{C_p T_{ti}}\right)$$

$$\boxed{\Phi = \frac{1}{\sqrt{2k}}\left(2k + \left(\frac{\gamma-1}{\gamma}\right)H\right)}$$

In a typical propulsion system, we want τ & $\dot{m}$ to be $\approx \phi$, So $\Delta\Phi = \phi$

$$\Phi = \frac{T}{\dot{m}(C_p T_{ti})^{1/2}} = \frac{p(1+\gamma M^2)}{\dot{m}(C_p T_{ti})^{1/2}}$$

$$\Rightarrow \Phi \propto \frac{M^2}{\dot{m}}$$

$\Rightarrow$ For turboprop (low Mach, high $\dot{m}$), $\Rightarrow \Phi \ll 1$

For turbofan (higher M, high $\dot{m}$), $\Rightarrow \Phi < 1$

For turbojets (higher M, lower $\dot{m}$) $\Rightarrow \Phi \leq 1$

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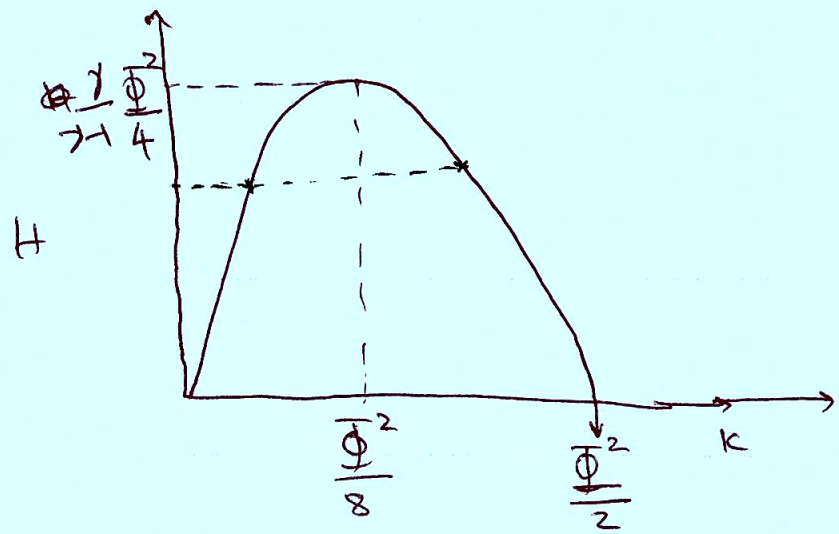
For rockets (lowest $\dot{m}$, highest M) $\Rightarrow \Phi > 1$

We will see that for ram/scram, $\Phi \approx 1$

Recall $\Phi = \frac{1}{\sqrt{2k}}\left(2k + \frac{\gamma-1}{\gamma}H\right)$

Q: What does const Φ look like in H-k ?



In H-k space, isocontours of $\Phi = \phi$ are asymmetric parabolas.

For every given value of H, there are 2 sol^{n}s for k (except for max). One side is subsonic & the other is supersonic.

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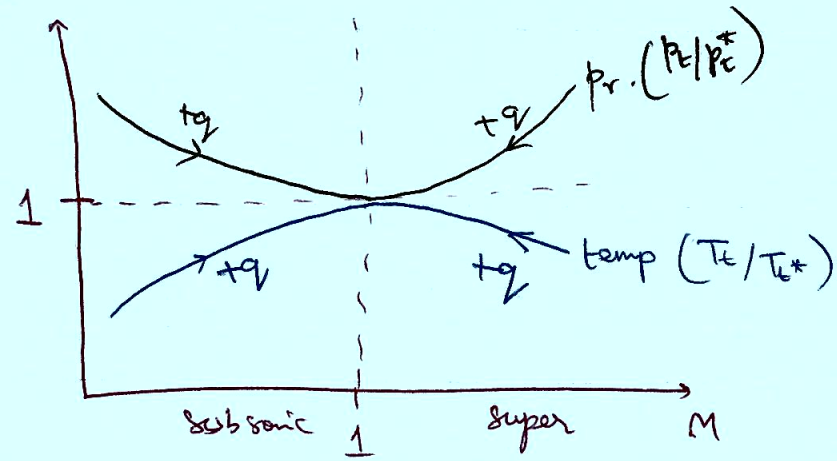
Recall our assumptions while deriving Φ below.

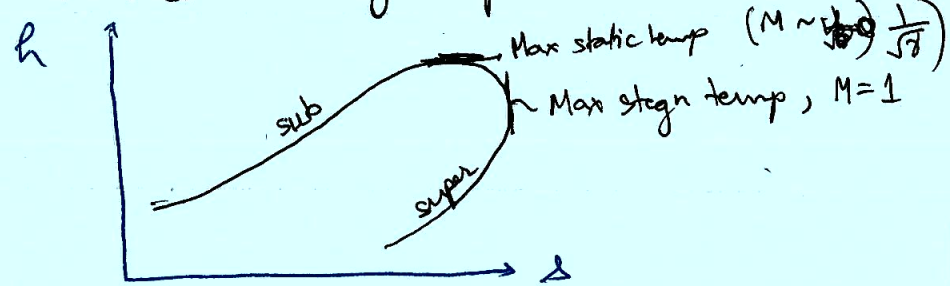
- 1D flow
- frictionless
- <u>have</u> heat add^n.
- ~~Const area~~ If we add constant area assumption, then we get ~~Rab~~ Rayleigh flow.

Recall results from Rayleigh flow

$\frac{T_t}{T_t^*}, \frac{p_t}{p_t^*}$



- As we add heat, we increase the stag. temp & drive the Mach# to 1. p_t always decreases.



- Max stag temperature occurs @ M=1
- Max static temp occurs @ M<1

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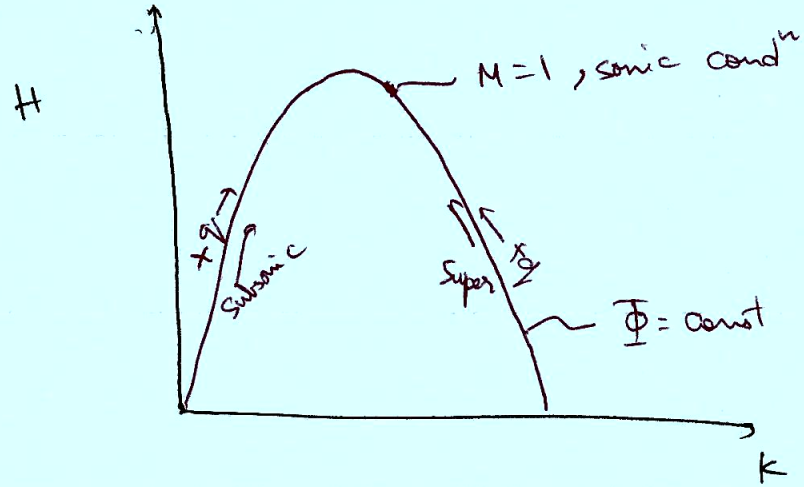
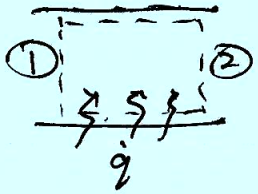


Recall $I = p + \rho u^2$

What happens when you add heat to the mom. eq^n

$$\underbrace{p_1 + \rho_1 u_1^2}_{I_1} = \underbrace{p_2 + \rho_2 u^2}_{I_2}$$



$\Rightarrow \Phi_{in} = \Phi_e \Rightarrow$

Rayleigh line $\Rightarrow \Phi = \text{const.}$

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Normal shocks in H-k space.

- For ramjets, need low subsonic flow into combustor
- Hence, we need normal shock to get flow from supersonic to subsonic.

What do contours of constant M# look like?

$$M = \frac{V}{a}$$

$$\Rightarrow M^2 = \frac{V^2}{a^2} = \frac{V^2}{\gamma R T}$$

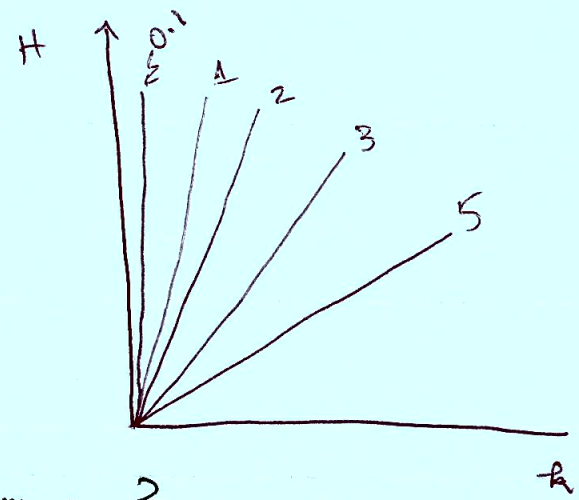
Note $R = \frac{(\gamma-1)C_p}{\gamma}$

$$\Rightarrow M^2 = \frac{V^2}{2C_pT_{ti}} \cdot \frac{2C_pT_{ti}}{\gamma R T}$$

$$M^2 = k \cdot \frac{2\,C_pT_{ti}}{(\gamma-1)C_pT}$$

$$M^2 = \frac{2k}{(\gamma-1)H}$$

So $H = \frac{2k}{(\gamma-1)M^2}$



$\Rightarrow$ Straight lines emanating from origin w/ slope $= \frac{2}{(\gamma-1)M^2}$

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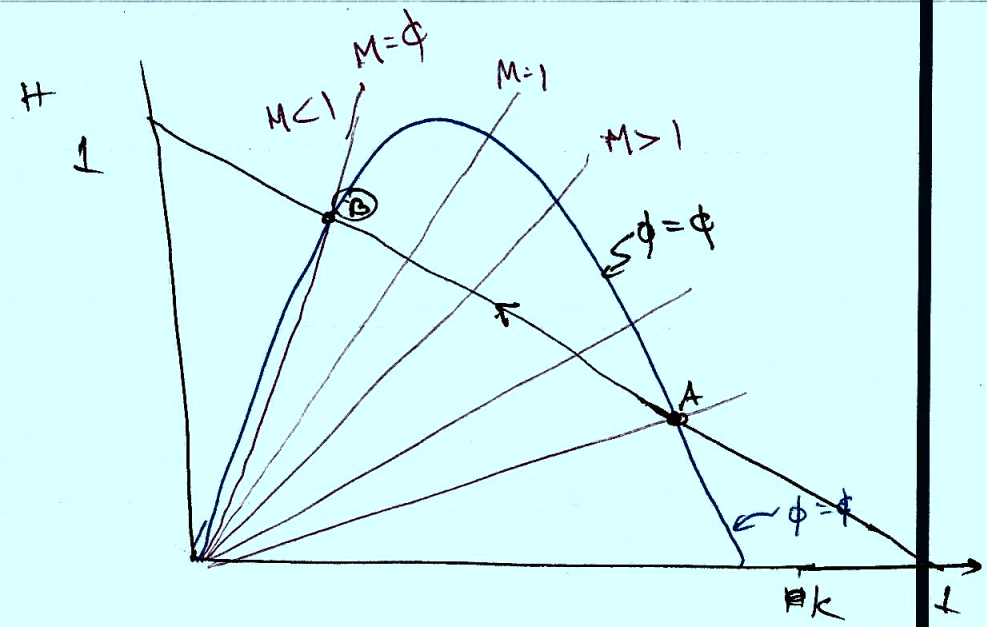
Across a normal shock

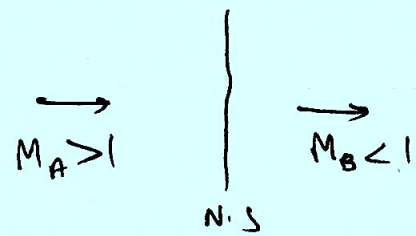
- Adiabatic.
- $M_1 > 1$; $M_2 < 1$.
- Φ = const across shock (no change in area or T_t)

  $\Rightarrow$ $\tau \propto \Delta\Phi$ & $\tau = 0$ for a shock (no thrust is produced)

- Since adiabatic $H + k = 1$



Normal shock goes from pt A to pt B along $T_t = \phi$



$k_A > k_B$

$H_A < H_B$.

For frictionless, const area, adiabatic flow, Φ = const & $H + k = 1$.

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There are 3 possible choices for Φ.

$\Phi > \Phi_c$, there are 2 sol^n's, one subsonic, one supersonic

$\Phi = \Phi_c$, there is only one sol^n.

$\Phi < \Phi_c$, no sol^n.

Turns out that Φ_c is only a funct^n of γ

Can show that $\Phi_c = \left(2\left(1 - \frac{1}{\gamma^2}\right)\right)^{1/2}$.

For $\gamma = 1.4$, $\Phi_c = 0.99$.

Recall $\dfrac{\tau}{\dot{m}(C_p T_{t_i})^{1/2}} = \Phi_2 - \Phi_1$

As we saw in Rayleigh flow (const area) adding heat does not change Φ, but for velocity changes due to area changes, Φ <u>will</u> change.

- Note for adiabatic flow, w/ (or without) area change, we can derive a relationship between M (which is a funct^n of area change) & Φ

$$\Phi = \left(\frac{(\gamma-1)M^2}{1+\frac{\gamma-1}{2}M^2}\right)^{1/2}\left(1+\frac{1}{\gamma M^2}\right)$$

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Overall,

Decelerating a supersonic flow adiabatically, reduces Φ

Accel. a supersonic flow adiabatically, increases Φ

Subsonic flow has opposite behavior.

~~Decel~~ Decel. a sub adiabatically, increase Φ

Accel " " " , decreases Φ.

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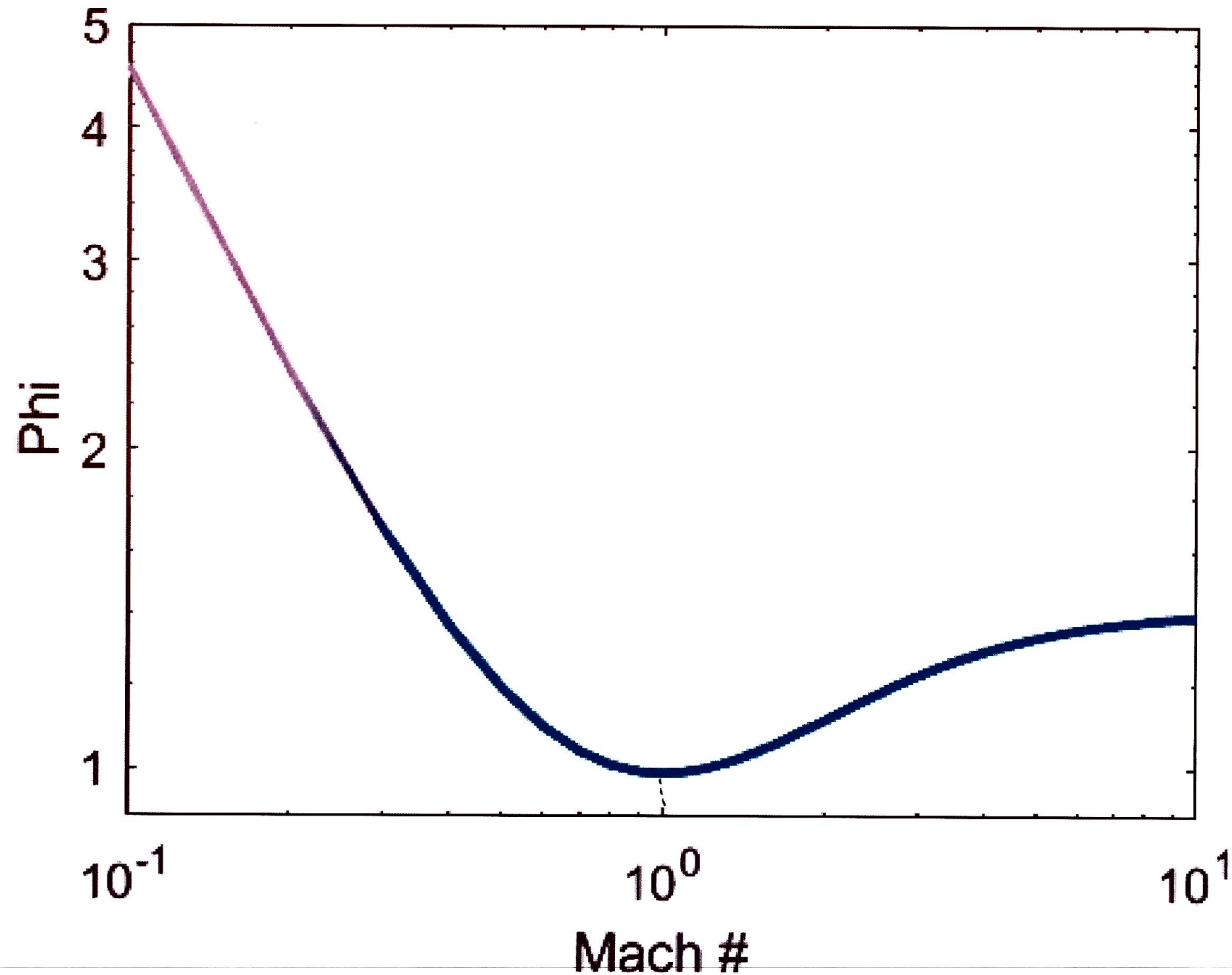
5
4
3
2
1
Phi
10^{-1}
10^{0}
10^{1}
Mach #

Some derivations.

- k as of (Φ) across normal shock (adiabatic)

$$\Phi = \frac{1}{\sqrt{2k}}\left(2k + \frac{\gamma-1}{\gamma}H\right) \rightarrow \text{(A)}$$

$$H + k = 1 \Rightarrow \Phi = \frac{1}{\sqrt{2k}}\left(2k - \left(\frac{\gamma-1}{\gamma}\right)k + \frac{\gamma-1}{\gamma}\right)$$

$$\gamma\sqrt{2k}\,\Phi = \left((\gamma+1)k + (\gamma-1)\right)$$

Let $\sqrt{k} = x$

$$\Rightarrow \sqrt{2}\,\gamma x\,\Phi = \left((\gamma+1)x^2 + (\gamma-1)\right)$$

$$\Rightarrow (\gamma+1)x^2 - \sqrt{2}\,\gamma x\,\Phi + (\gamma-1) = 0$$

$$\Rightarrow x = \frac{\sqrt{2}\,\gamma\Phi \pm \sqrt{2\gamma^2\Phi^2 - 4(\gamma^2-1)}}{2(\gamma+1)}$$

$$\Rightarrow \boxed{k = x^2 = \left\{\frac{\gamma\Phi}{\sqrt{2}(\gamma+1)} \pm \sqrt{\frac{\gamma\Phi^2}{2(\gamma+1)^2} - \left(\frac{\gamma-1}{\gamma+1}\right)}\right\}^2} \quad (1)$$

- For $\Phi = \phi$ to have single value in h-k diag., need only one sol^n for (1)

$$\Rightarrow \sqrt{\frac{\gamma\Phi_c^2}{2(\gamma+1)^2} - \left(\frac{\gamma-1}{\gamma+1}\right)} = 0 \Rightarrow \boxed{\Phi_c = \sqrt{2\left(1 - \frac{1}{\gamma^2}\right)}} \quad (2)$$

$$\boxed{\Phi_c = 0.99} \text{ for } \boxed{\gamma = 1.4} \quad (3)$$

$$\Rightarrow \boxed{\frac{V_c^2}{2c_pT_{tc}} = \frac{\gamma-1}{\gamma+1}} \quad (4) \Rightarrow \boxed{\frac{V_c^2}{2c_pT_{tc}} = 0.167 \text{ @ } \gamma = 1.4} \Rightarrow M_c = ?$$

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□ 1D - heat addition problem (Rayleigh flow)

Let $\frac{T_e}{T_i} = \tau_e = H_e + k_e$ @ exit.

Subst for $H_e = \tau_e - k_e$ in eqn (A)

$$\Rightarrow k_e = \left\{ \frac{\gamma \Phi_i}{(\gamma+1)\sqrt{2}} \pm \sqrt{\frac{1}{2}\left(\frac{\gamma \Phi_i}{\gamma+1}\right)^2 - \tau_e\left(\frac{\gamma-1}{\gamma+1}\right)} \right\}^2 \quad (5)$$

For single value of k_e

$$\boxed{\tau_e = \frac{\Phi_i^2}{2\left(1 - \frac{1}{\gamma^2}\right)}} \quad (6)$$

Eg. $\tau_e = 1.47$ for $\Phi_i = 1.2$ & $\gamma = 1.4$.

$$\boxed{k_e = \frac{V_e^2}{2 C_p T_{ti}} = \frac{1}{2}\left(\frac{\gamma \Phi_i}{\gamma+1}\right)^2} \quad (7)$$

□ What is $M_e = ?$

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□ Φ as $f(M)$ in adiabatic flow. (w/ area change or without)

$$\Phi = \frac{1}{\sqrt{2k}}\left(2k + \left(\frac{\gamma-1}{\gamma}\right)H\right)$$

Adiabatic flow $\Rightarrow$ $H + k = 1 \Rightarrow H = 1 - k$.

$$M^2 = u^2/a^2 = u^2/\gamma RT = \frac{u^2}{c_p T_{ti}} \cdot \frac{c_p T_{ti}}{\gamma RT}$$

$$M^2 = 2k \cdot \frac{c_p T_{ti}}{(\gamma-1) c_p T} \qquad \left(R = \frac{(\gamma-1)}{\gamma} c_p\right)$$

$$\Rightarrow M^2 = \frac{2k}{(\gamma-1)} \cdot \frac{1}{H} \quad \text{or} \quad H = \frac{2k}{M^2} \cdot \frac{1}{\gamma-1}$$

$$\Rightarrow \Phi = \frac{1}{\sqrt{2k}}\left(2k + \frac{2k}{M^2(\gamma-1)} \cdot \frac{\gamma-1}{\gamma}\right)$$

$$= \sqrt{2k}\left(1 + \frac{1}{\gamma M^2}\right)$$

$$2k = \frac{u^2}{c_p T_{ti}} = \frac{u^2}{\gamma RT} \cdot \frac{\gamma RT}{c_p T_{ti}} = M^2 \cdot \frac{(\gamma-1)T}{T_{ti}} = \frac{(\gamma-1)M^2}{\left(1 + \frac{\gamma-1}{2}M^2\right)}$$

$$\Rightarrow \boxed{\Phi = \sqrt{\frac{(\gamma-1)M^2}{\left(1+\frac{\gamma-1}{2}M^2\right)}} \cdot \left(1 + \frac{1}{\gamma M^2}\right)}$$

□ For flow w/ heat add^n: $\tau_e = \frac{T_{te}}{T_{ti}} \Rightarrow \Phi = \left[\left(\frac{\tau_e (\gamma-1) M^2}{1+\left(\frac{\gamma-1}{2}\right)M^2}\right)^{1/2} \cdot \left(1 + \frac{1}{\gamma M^2}\right)\right]$

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